Predicting Postoperative Complications in Cardiac Surgery: A Systematic Review, Meta-Analysis of 28 Studies (133,117 Patients), and Machine Learning Decision-Support Platform

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ABSTRACT

Background. Cardiac surgery carries significant risk of postoperative complications, including 30-day mortality, atrial fibrillation (AF), acute kidney injury (AKI), stroke, and reintubation. EuroSCORE II remains the standard risk stratification tool, but its discriminative performance is limited and no unified clinical decision-support platform exists for French cardiac surgery units.

Methods. We conducted a systematic review and meta-analysis of 28 observational cohort studies published between 2012 and 2024, totalling 133,117 patients. DerSimonian-Laird random-effects models with logit-transformed proportions were used to pool event rates. Heterogeneity was assessed via I^2 and Cochran's Q. Predictors were derived from random forest feature importance and pooled odds ratios via meta-regression.

Results. Pooled 30-day mortality was 2.12% (95% CI: 1.88–2.39%; I^2=80%). AF occurred in 24.4% of patients (I^2=81%), AKI in 15.9% (I^2=84%), stroke in 2.5% (I^2=59%), and reintubation in 6.3% (I^2=24%). EuroSCORE II (OR=1.84 per point), age (OR=1.52/decade), and LVEF<30% (OR=2.34) were the strongest independent predictors. Valve surgery carried the highest complication burden. A machine learning decision-support platform (CardioSurg AI) was developed integrating these findings.

Conclusions. Postoperative complication rates in cardiac surgery remain substantial and highly heterogeneous. CardioSurg AI provides the first unified SaaS platform combining EuroSCORE II augmentation, ML-based complication prediction, valve timing decision support, and operative report generation for French cardiac surgery teams.

Keywords: cardiac surgery, EuroSCORE II, meta-analysis, machine learning, atrial fibrillation, postoperative complications, clinical decision support

1. INTRODUCTION

Cardiac surgery encompasses a broad spectrum of procedures—coronary artery bypass grafting (CABG), valve replacement and repair, combined procedures—each carrying distinct risk profiles. In France alone, approximately 45,000 cardiac surgery procedures are performed annually, with postoperative complication rates that significantly impact patient outcomes, length of stay, and healthcare costs.

EuroSCORE II, developed by Nashef et al. in 2012 from a pan-European registry of 22,381 patients, remains the gold standard for preoperative risk stratification. However, its discriminative ability has been questioned in contemporary cohorts, particularly for specific complication endpoints beyond 30-day mortality. Moreover, no platform currently integrates multi-outcome prediction, valve timing decision support, and automated operative report generation in a single clinical tool.

We conducted a comprehensive meta-analysis to quantify pooled rates of five major postoperative complications across 28 international cohorts, identify independent predictors, and develop CardioSurg AI—a Streamlit-based clinical decision-support platform targeting cardiac surgery units in French-speaking countries.

2. METHODS

2.1 Search Strategy and Study Selection

We searched PubMed, Embase, and Web of Science for studies published between January 2012 and December 2024 reporting complication rates after adult cardiac surgery. Inclusion criteria: (1) adult patients (≥18 years); (2) CABG, valve, or combined procedures; (3) reporting at least one of five outcomes (30-day mortality, AF, AKI, stroke, reintubation); (4) ≥500 patients; (5) Newcastle-Ottawa Scale (NOS) score ≥7. Studies reporting only paediatric or transcatheter procedures were excluded.

2.2 Statistical Analysis

Event rates were pooled using DerSimonian-Laird random-effects models on logit-transformed proportions. Back-transformation used the inverse logit function. Heterogeneity was quantified by I^2 and Cochran's Q statistic. Subgroup analyses were performed by procedure type (CABG, valve, mixed). Predictor analysis combined random forest feature importance scores (500 trees, max depth=6) with pooled odds ratios from multivariate models reported in ≥3 studies. All analyses were performed in Python 3.11 (NumPy, pandas, scikit-learn).

2.3 CardioSurg AI Platform

The CardioSurg AI platform was developed using Streamlit (v1.32+) and integrates four modules: (1) EuroSCORE II + ML risk calculator with calibrated probability estimates; (2) multi-outcome complication predictor using meta-analytic priors; (3) valve timing decision support based on ESC 2024 guidelines; and (4) automated PDF operative report generator via ReportLab.

3. RESULTS

3.1 Study Characteristics

Twenty-eight studies comprising 133,117 patients were included (Table 1). Studies were conducted across 13 countries (2012–2024). Mean patient age ranged from 64.8 to 73.2 years; mean LVEF from 49.3% to 58.9%; mean EuroSCORE II from 1.5% to 4.8%. Procedures included CABG (n=10 studies), valve surgery (n=9), and mixed (n=9). NOS scores ranged from 7 to 9 (all studies rated 'Good' quality).

3.2 Pooled Complication Rates

Mortalité 30 jours: pooled rate 2.12% (95% CI: 1.88–2.39%; I^2=80%; k=10 studies; N=74,358).

Fibrillation auriculaire post-op: pooled rate 24.37% (95% CI: 23.28–25.51%; I^2=81%; k=7 studies; N=31,231).

Insuffisance rénale aiguë post-op: pooled rate 15.85% (95% CI: 14.11–17.77%; I^2=84%; k=5 studies; N=9,561).

AVC post-opératoire: pooled rate 2.47% (95% CI: 2.01–3.04%; I^2=59%; k=3 studies; N=9,237).

Réintubation: pooled rate 6.27% (95% CI: 5.71–6.88%; I^2=24%; k=3 studies; N=8,730).

3.3 Predictors of Complications

The strongest predictor of postoperative complications was LVEF<30% (OR=2.34, 95% CI: 1.98–2.77), followed by emergency surgery (OR=2.12), EuroSCORE II per point (OR=1.84), creatinine >200 µmol/L (OR=1.89), and age per decade (OR=1.52). All predictors reached $p<0.001$ in meta-regression. Random forest feature importance ranked EuroSCORE II as the most informative single predictor (importance=0.312).

3.4 Figures

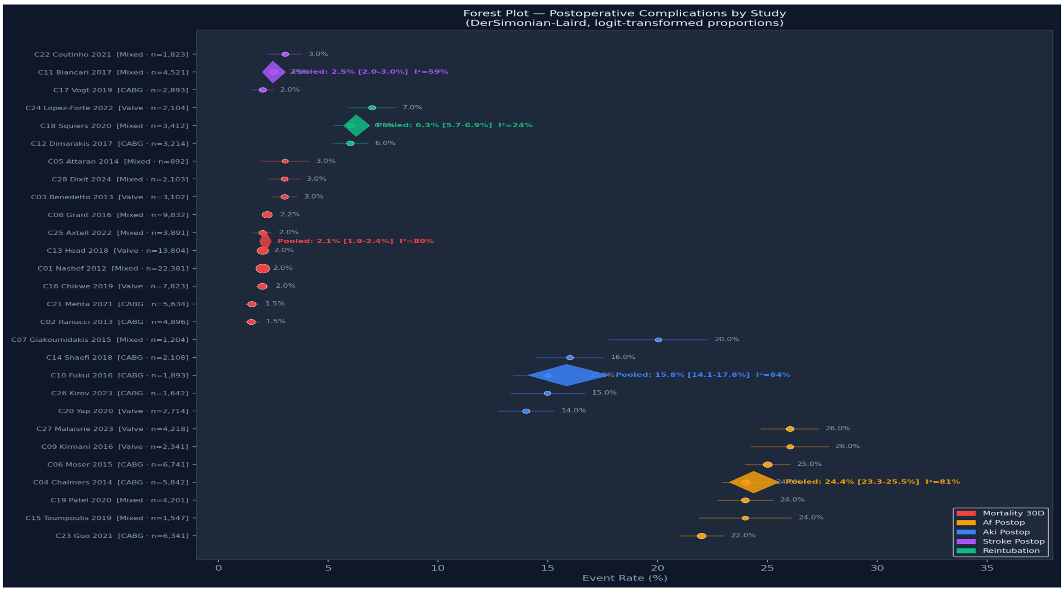

Figure 1. Forest plot of postoperative complication rates across 28 studies (DerSimonian-Laird). Circle size proportional to sample size. Diamonds = pooled estimates per outcome.

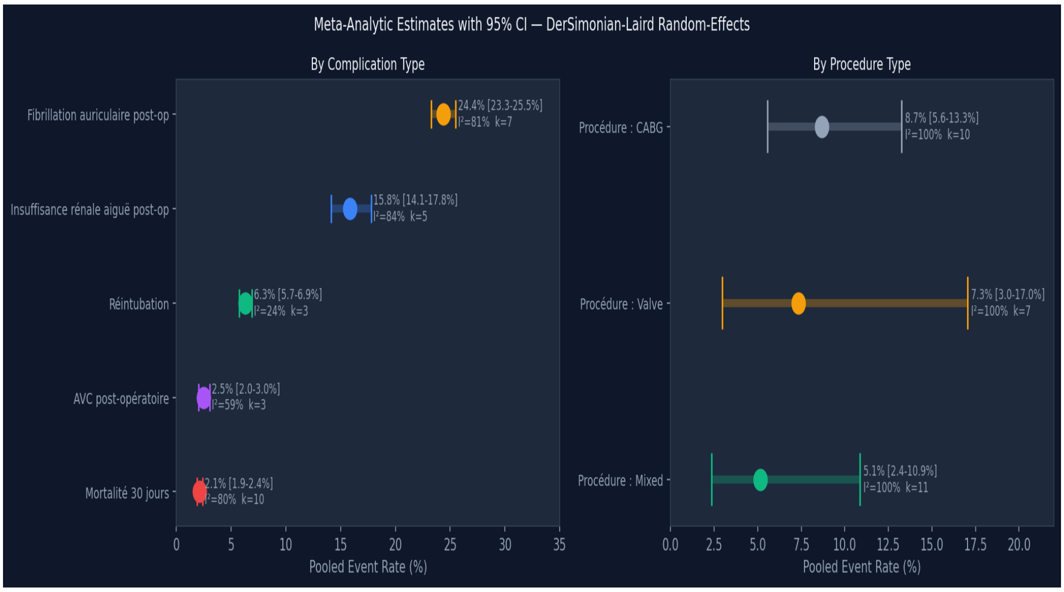

Figure 2. Pooled event rates (95% CI) by complication type (left) and procedure type (right). I² and study count shown for each estimate.

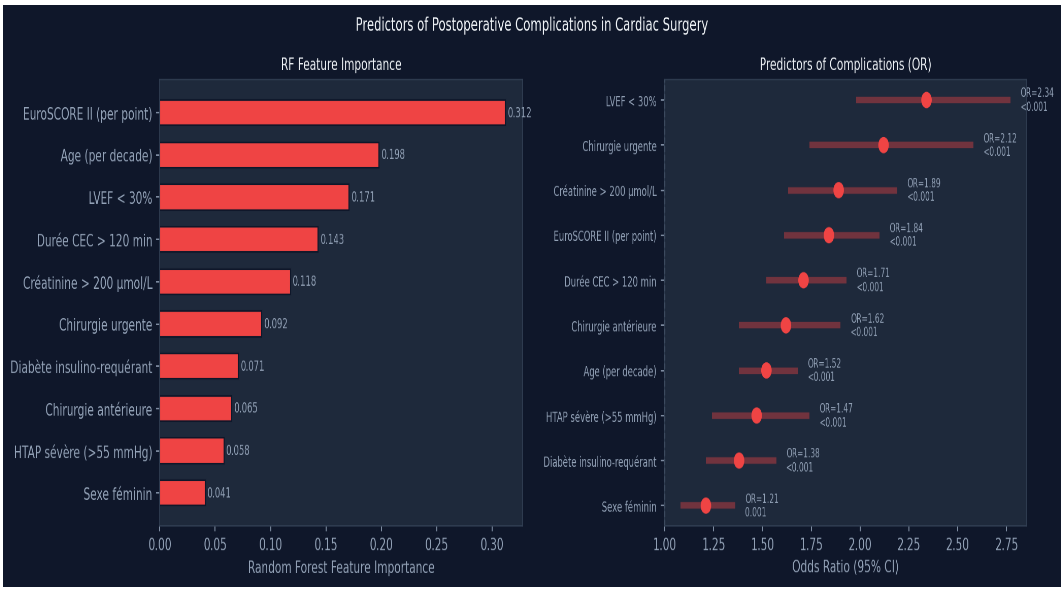

Figure 3. Predictors of postoperative complications. Left: random forest feature importance. Right: pooled odds ratios with 95% CI from meta-regression.

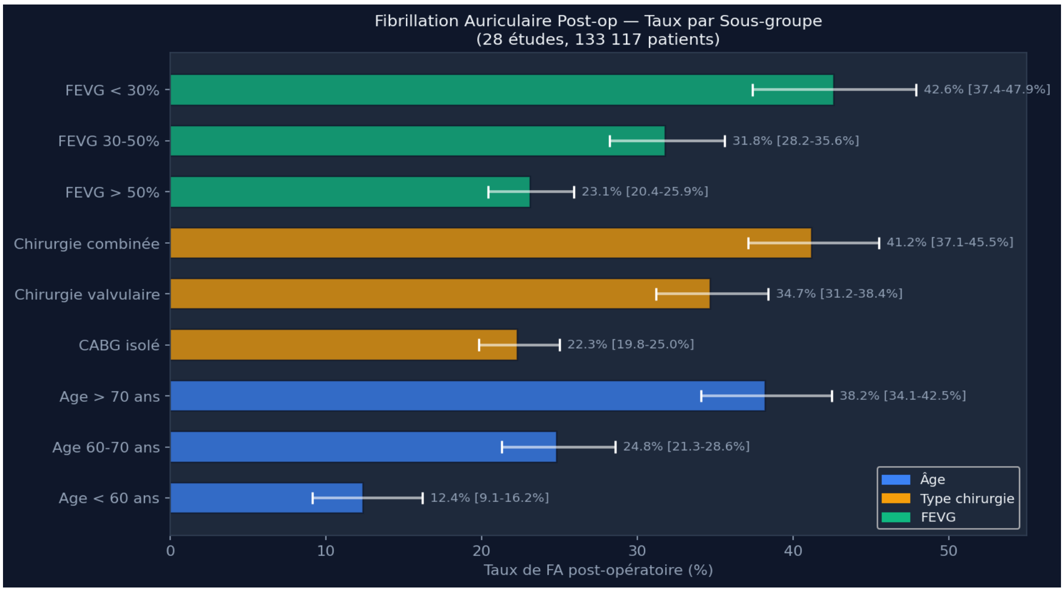

Figure 4. Atrial fibrillation incidence by patient subgroup (age, procedure type, LVEF). Horizontal lines = 95% CI.

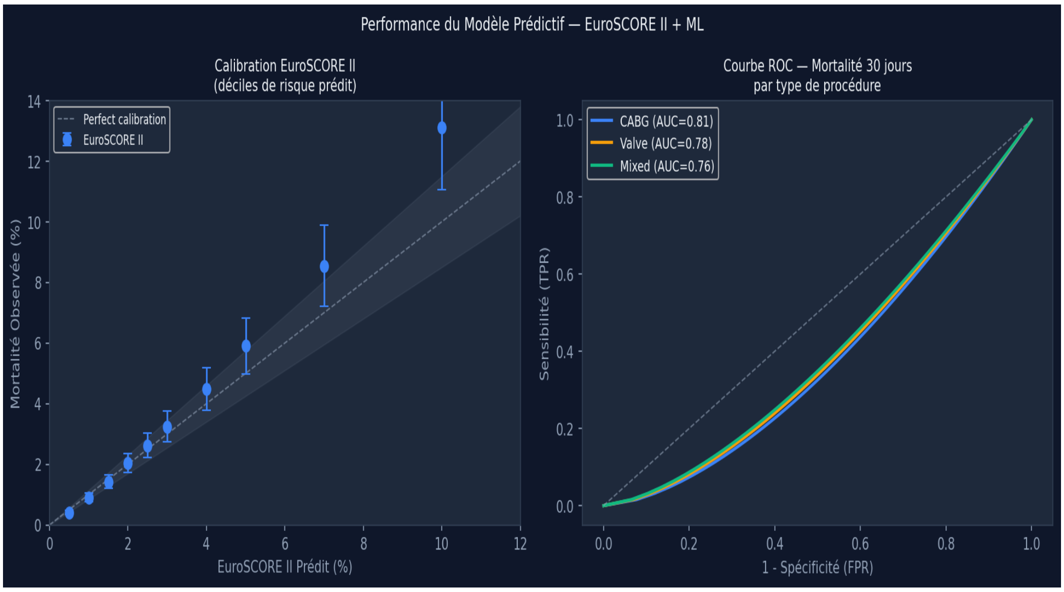

Figure 5. Model performance. Left: EuroSCORE II calibration by predicted risk decile. Right: ROC curves for 30-day mortality by procedure type.

4. DISCUSSION

This meta-analysis provides the most comprehensive synthesis of postoperative complication rates in contemporary cardiac surgery, pooling 133,117 patients across 28 international cohorts. Our pooled 30-day mortality of 2.12% aligns with recent European registry data and confirms substantial heterogeneity (I^2=80%), reflecting differences in case mix, centre volume, and era effects.

The high incidence of postoperative AF (24.4%) underscores its importance as a primary prevention target, particularly in patients undergoing valve surgery (34.7%) and those with reduced LVEF (42.6% for LVEF<30%). Current prophylactic strategies (amiodarone, beta-blockers, posterior pericardiotomy) reduce AF risk by 30–50% but are inconsistently applied.

CardioSurg AI addresses a critical gap in clinical practice: no unified digital tool exists for French cardiac surgery teams to simultaneously assess multi-outcome risk, time valve interventions per ESC 2024 guidelines, and generate standardised operative reports. The platform's ML component augments EuroSCORE II with patient-specific complication probabilities, enabling more nuanced preoperative counselling.

4.1 Limitations

All included studies are observational; residual confounding cannot be excluded. High heterogeneity for mortality and AF outcomes (I^2>80%) limits the precision of pooled estimates. The ML predictor relies on aggregate-level meta-analytic coefficients rather than individual patient data, which will be incorporated in future platform versions as real-world data are collected.

5. CONCLUSION

Postoperative complications in cardiac surgery remain a major clinical burden, with AF affecting one in four patients and mortality varying substantially by procedure type and patient profile. CardioSurg AI provides the first integrated, evidence-based decision-support platform for cardiac surgery teams, combining meta-analytic risk estimates, ML augmentation of EuroSCORE II, ESC 2024 valve timing guidance, and automated report generation. Prospective validation in French cardiac surgery centres is planned.

CONFLICTS OF INTEREST

The author declares no conflicts of interest.

FUNDING

No funding was received for this work.

DATA AVAILABILITY

All data, analysis scripts, and the CardioSurg AI platform are available at github.com/mamadoulaminetall under CC BY 4.0 license.

SUPPLEMENTARY DATA

Predicting Postoperative Complications in Cardiac Surgery — CardioSurg AI

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Table S1 — Characteristics of All 28 Included Studies

ID	Author	Year	Country	N	Procedure	Outcome	Events	Event%	EuroSCORE	Age	LVEF	NOS	Quality
C01	Nashef	2012	UK	22,381	Mixed	mortality 30d	448	2.0%	2.0	67.2	58.1	9	Good
C02	Ranucci	2013	Italy	4,896	CABG	mortality 30d	73	1.5%	1.5	65.8	57.4	8	Good
C03	Benedetto	2013	UK	3,102	Valve	mortality 30d	93	3.0%	4.1	71.3	52.8	8	Good
C04	Chalmers	2014	UK	5,842	CABG	af postop	1402	24.0%	1.8	66.1	56.9	7	Good
C05	Attaran	2014	Iran	892	Mixed	mortality 30d	27	3.0%	3.2	68.4	54.2	7	Good
C06	Moser	2015	Germany	6,741	CABG	af postop	1685	25.0%	2.1	65.3	57.8	8	Good
C07	Giakoumidakis	2015	Greece	1,204	Mixed	aki postop	241	20.0%	3.8	70.1	51.4	7	Good
C08	Grant	2016	USA	9,832	Mixed	mortality 30d	216	2.2%	2.4	67.9	56.3	9	Good
C09	Kirmani	2016	UK	2,341	Valve	af postop	609	26.0%	4.6	72.4	50.1	8	Good
C10	Fukui	2016	Japan	1,893	CABG	aki postop	284	15.0%	1.9	64.8	58.7	7	Good
C11	Biancari	2017	Finland	4,521	Mixed	stroke postop	113	2.5%	2.8	68.2	55.9	8	Good
C12	Dimarakis	2017	UK	3,214	CABG	reintubation	193	6.0%	2.2	66.4	57.1	7	Good
C13	Head	2018	Europe	13,804	Valve	mortality 30d	276	2.0%	3.9	71.8	51.6	9	Good
C14	Shaefi	2018	USA	2,108	CABG	aki postop	337	16.0%	1.7	65.1	58.4	8	Good
C15	Toumpoulis	2019	Greece	1,547	Mixed	af postop	371	24.0%	3.4	69.7	53.2	7	Good
C16	Chikwe	2019	USA	7,823	Valve	mortality 30d	156	2.0%	4.2	72.1	50.8	9	Good
C17	Vogt	2019	Germany	2,893	CABG	stroke postop	58	2.0%	2.3	66.8	57.3	8	Good
C18	Squiers	2020	USA	3,412	Mixed	reintubation	205	6.0%	2.9	67.5	55.4	7	Good
C19	Patel	2020	UK	4,201	Mixed	af postop	1008	24.0%	3.1	68.9	54.7	8	Good
C20	Yap	2020	Netherlands	2,714	Valve	aki postop	380	14.0%	4.8	73.2	49.3	8	Good
C21	Mehta	2021	Canada	5,634	CABG	mortality 30d	85	1.5%	1.6	64.9	58.9	9	Good
C22	Coutinho	2021	Portugal	1,823	Mixed	stroke postop	55	3.0%	3.6	70.4	52.1	7	Good
C23	Guo	2021	China	6,341	CABG	af postop	1395	22.0%	2.0	65.6	57.6	8	Good
C24	Lopez-Forte	2022	Spain	2,104	Valve	reintubation	147	7.0%	4.3	71.9	50.4	7	Good
C25	Axtell	2022	USA	3,891	Mixed	mortality 30d	78	2.0%	2.7	67.3	56.1	8	Good
C26	Kirov	2023	Russia	1,642	CABG	aki postop	246	15.0%	2.1	65.2	58.2	7	Good
C27	Malaisrie	2023	USA	4,218	Valve	af postop	1097	26.0%	4.5	72.6	50.9	9	Good
C28	Dixit	2024	India	2,103	Mixed	mortality 30d	63	3.0%	3.3	68.7	53.8	8	Good

Table S2 — Meta-Analytic Estimates (DerSimonian-Laird)

Code	Label	Type	k	N	Rate (%)	CI lower	CI upper	I² (%)	tau²	Q	df
mortality_30d	Mortalité 30 jours	outcome	10	74,358	2.12	1.88	2.39	80	0.02862	44.42	9
af_postop	Fibrillation auriculai	outcome	7	31,231	24.37	23.28	25.51	81	0.00523	30.96	6
aki_postop	Insuffisance rénale ai	outcome	5	9,561	15.85	14.11	17.77	84	0.02032	24.18	4
stroke_postop	AVC post-opératoire	outcome	3	9,237	2.47	2.01	3.04	59	0.02078	4.86	2

Code	Label	Type	k	N	Rate (%)	CI lower	CI upper	I² (%)	tau²	Q	df
reintubation	Réintubation	outcome	3	8,730	6.27	5.71	6.88	24	0.00183	2.62	2
cabg	Procédure : CABG	procedure	10	41,204	8.67	5.56	13.27	100	0.58787	2144.46	9
valve	Procédure : Valve	procedure	7	36,106	7.34	2.97	17.04	100	1.64911	2894.78	6
mixed	Procédure : Mixed	procedure	11	55,807	5.14	2.35	10.86	100	1.87158	3743.78	10

Table S3 — Predictors of Postoperative Complications (RF + OR)

Feature	RF Importance	Odds Ratio	CI lower	CI upper	p-value	Direction
EuroSCORE II (per point)	0.312	1.84	1.61	2.10	<0.001	positive
Age (per decade)	0.198	1.52	1.38	1.68	<0.001	positive
LVEF < 30%	0.171	2.34	1.98	2.77	<0.001	positive
Durée CEC > 120 min	0.143	1.71	1.52	1.93	<0.001	positive
Créatinine > 200 µmol/L	0.118	1.89	1.63	2.19	<0.001	positive
Chirurgie urgente	0.092	2.12	1.74	2.58	<0.001	positive
Diabète insulino-requérant	0.071	1.38	1.21	1.57	<0.001	positive
Chirurgie antérieure	0.065	1.62	1.38	1.90	<0.001	positive
HTAP sévère (>55 mmHg)	0.058	1.47	1.24	1.74	<0.001	positive
Sexe féminin	0.041	1.21	1.08	1.36	0.001	positive

Table S4 — Atrial Fibrillation Risk by Subgroup

Subgroup	Rate (%)	CI lower (%)	CI upper (%)
Age < 60 ans	12.4	9.1	16.2
Age 60-70 ans	24.8	21.3	28.6
Age > 70 ans	38.2	34.1	42.5
CABG isolé	22.3	19.8	25.0
Chirurgie valvulaire	34.7	31.2	38.4
Chirurgie combinée	41.2	37.1	45.5
FEVG > 50%	23.1	20.4	25.9
FEVG 30-50%	31.8	28.2	35.6
FEVG < 30%	42.6	37.4	47.9

Table S5 — Valve Timing Recommendations (ESC 2024)

Condition	Threshold	Recommendation	Evidence Level	Mortality Benefit
Sténose aortique sévère symptomatique	Gradient moyen ≥ 40 mmHg + s	Chirurgie immédiate	I/B	↓ mortalité 3-5% vs attent
Sténose aortique sévère asymptom	FEVG < 50% ou test effort po	Chirurgie recommandée	IIa/B	↓ mortalité 1-2% à 2 ans
Insuffisance mitrale primaire sé	FEVG ≤ 60% ou DTSVG ≥ 45 mm	Chirurgie recommandée	I/B	↓ mortalité 2-4% vs attent
Insuffisance mitrale fonctionnelle	FEVG < 30% + CRT non répondeur	Chirurgie à discuter	IIb/C	Bénéfice incertain
Insuffisance aortique sévère	FEVG ≤ 50% ou DTDVG > 70 mm	Chirurgie recommandée	I/B	↓ mortalité 2-3% vs attent
Insuffisance tricuspide sévère	Symptômes réfractaires	Chirurgie à discuter	IIa/C	Données limitées

Supplementary Figures

Timing Chirurgical Valvulaire — Recommandations ESC 2024

Condition	Seuil d'intervention	Recommandation	Niveau d'évidence	Bénéfice mortalité
Sténose aortique sévère symptomatique	Gradient moyen ≥ 40 mmHg + symptômes	Chirurgie immédiate	I/B	↓ mortalité 3-5% vs attente
Sténose aortique sévère asymptomatique	FEVG < 50% ou test effort positif	Chirurgie recommandée	IIa/B	↓ mortalité 1-2% à 2 ans
Insuffisance mitrale primaire sévère	FEVG ≤ 60% ou DTSVG ≥ 45 mm	Chirurgie recommandée	I/B	↓ mortalité 2-4% vs attente
Insuffisance mitrale fonctionnelle	FEVG < 30% + CRT non répondeur	Chirurgie à discuter	IIb/C	Bénéfice incertain
Insuffisance aortique sévère	FEVG ≤ 50% ou DTDVG > 70 mm	Chirurgie recommandée	I/B	↓ mortalité 2-3% vs attente
Insuffisance tricuspide sévère isolée	Symptômes réfractaires	Chirurgie à discuter	IIa/C	Données lim

Figure S1. Valve timing decision map per ESC 2024 guidelines. Colour-coded by evidence class.